

Sudip Sharma

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EDUCATION

Master of Science in Advanced Computing, *Morgan State University*  May 2026 | Baltimore, US
Relevant Coursework: Design And Analysis Of Algorithms, Software Engineering, Cloud Computing

Bachelor of Science in Computer Science, *Kathmandu University*  2018 – 2023 | Dhulikhel, Nepal
Relevant Coursework: Data Structures and Algorithms, Algorithms and Complexity, Database Management Systems

PROFESSIONAL EXPERIENCE

Center for Equitable Artificial Intelligence and Machine Learning Systems , May 2025 – Aug 2025 | Baltimore, US
Graduate Mentor (Morgan State University)

- Led a team of undergraduates on multimodal deep learning pipelines combining 1D-CNN, Transformer, and 2D-CNN architectures for ECG classification, boosting overall model accuracy to 90%+
- Designed a feature extraction framework across three domains: time, frequency, and time-frequency using **Python**, **NumPy**, and **TensorFlow**, which increased ensemble performance by 14%
- Mentored students on data preprocessing pipelines (denoising, segmentation, augmentation) reducing model training time by 30% through optimized data loaders and GPU utilization

EB Pearls , *Software Engineer* Jul 2023 – Aug 2024 | Lalitpur, Nepal

- Designed and deployed secure authentication systems (login, OTP, token-based authentication) with **SwiftUI** and **JWT**, improving login success rates by 40% and user satisfaction by 63%
- Integrated Google Calendar API to sync events, improving scheduling efficiency and engagement for 1K+ active users
- Developed high-performance **GraphQL** APIs in **NestJS**, enhancing data retrieval speed by 37% across mobile and admin platforms
- Collaborated with cross-functional teams to deploy production-ready features, reducing bug reports by 25% through rigorous testing and code reviews

EB Pearls , *Software Engineering Intern* Apr 2023 – Jul 2023 | Lalitpur, Nepal

- Developed iOS mobile apps using **SwiftUI** and **UIKit**, implementing scalable design patterns (MVVM) and boosting in-app session length by 18% through intuitive navigation and animations
- Built and optimized **RESTful** and **GraphQL** backend services using **Node.js** and **Express**, improving API response time by 22% via load balancing and query optimization
- Prototyped and integrated real-time notification systems using **WebSockets**, enhancing user engagement for core product features by 13%

PROJECTS & LEADERSHIP

PillSenseAI - HopHacks 2025 , *Johns Hopkins University* Baltimore, US

- Built a full-stack medication safety assistant in a **React** + **Express**/**MongoDB** monorepo by implementing OCR-based prescription parsing (**Tesseract.js**) and **JWT**-secured APIs, enabling structured extraction of drug, dose, and frequency data from raw label images for end-to-end user workflows
- Improved parsing accuracy and system reliability by adding Gemini-based structured extraction, idempotent scan deduplication, and rules-based interaction/overdose checks, reducing duplicate entries and surfacing actionable safety flags

EventHub , *Morgan State University* Baltimore, US

- Architected and built EventHub, a full-stack campus event management platform using the **MERN** stack, by designing RESTful APIs, JWT-based authentication, and role-based access control, enabling secure event creation, discovery, and real-time management for students and organizers
- Optimized backend performance and scalability by structuring the project as a pnpm **monorepo** with modular **Express** services and efficient MongoDB query design, reducing API response latency and improving cross-team development velocity

Mastercard Center for Inclusive Growth , *AUC Data Science Initiative* Atlanta, US

- Secured **3rd place** by analyzing economic inclusion disparities across demographics using **Python**, **Pandas**, and **Scikit-learn** to build predictive models for financial inclusion metrics
- Visualized actionable insights using **Matplotlib** and **Seaborn**, helping policymakers understand disparities in digital banking and credit access

IT MEET 2022 , *Non-tech Events Lead* Dhulikhel, Nepal

- Led the planning and execution of 5+ events including Career Fair, Alumni Meet, Blood Donation Drive, Acoustic Night, and Idea Pitching Competitions, boosting student participation by 21%

SKILLS

Languages: Python, TypeScript, Swift | **Backend:** Node.js, NestJS, Express, GraphQL, REST APIs | **Frontend:** React, Swift UI, UIKit | **Databases:** MongoDB, MySQL | **Cloud:** AWS (EC2, S3, VPC, SageMaker)

AWARDS

Presidential Medal for Service and Excellence , *STAR Scholars Network* 16 Feb 2026

Awarded the Presidential Medal for Service and Excellence by the STAR Scholars Network at its 8th Anniversary Celebration in Washington, DC, for sustained leadership and advancing organizational growth and global impact as Media Specialist

PUBLICATIONS

Trust and Transparency: Making AI Explainable in Healthcare, *Springer Nature Switzerland (International)* (2026 Pending)

Interpretable Time-Series Deep Learning for Monitoring Disease Progression in Chest Radiography, *IEEE-EIT 2026* (Under Review)

A Time-Aware Clinical Decision Support System for Longitudinal Radiography Triage, *IEEE SMC 2026* (Under Review)